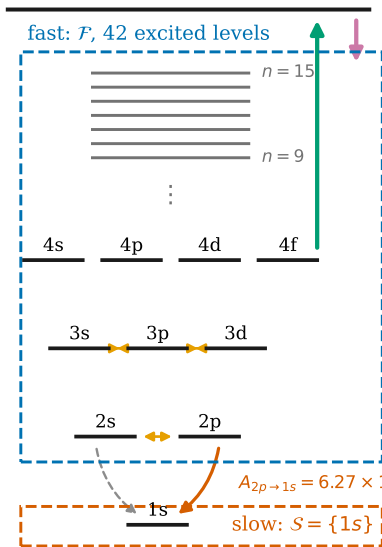


electron impact
excitation and
de-excitation



continuum ($\text{H}^+ + \text{e}^-$)

recombination, entering as
the source vector (every state)

$n = 9$ to 15 : bundled, 7 states

electron-impact ionisation
(every state)

$n = 5$ to 8 : ℓ -resolved, not drawn

$n \leq 8$: ℓ -resolved, 36 states

proton ℓ -mixing, $\Delta n = 0$
(35 states: every resolved $n \geq 2$)

2s has NO E1 decay to 1s ($A = 0$ exactly). It is still fast:
proton ℓ -mixing to 2p carries 99.97% of its loss rate at the
benchmark, and never less than 99.92% anywhere on the grid. That is
why $S = \{1s\}$ is the only slow state.

true energy scale

